

UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

Object Oriented Programming

36863

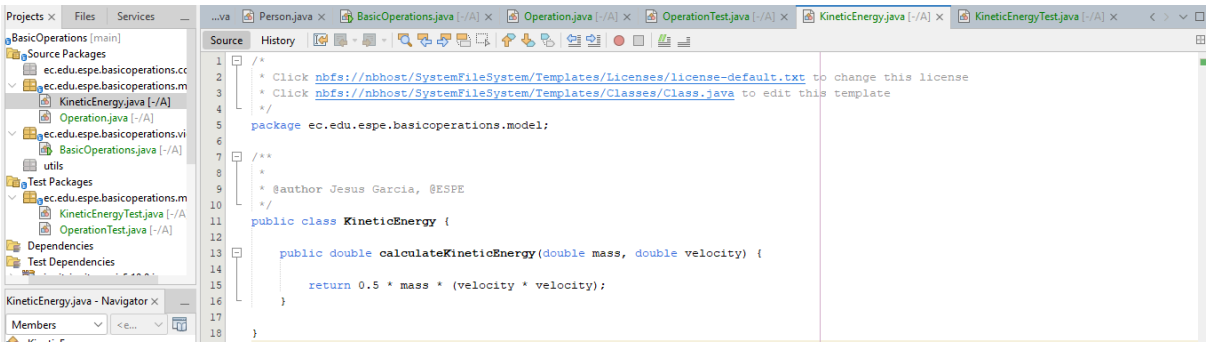
WorkShop16-LibsAndUnitTest

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Evidences

KineticEnergy Class



```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package ec.edu.espe.basicoperations.model;

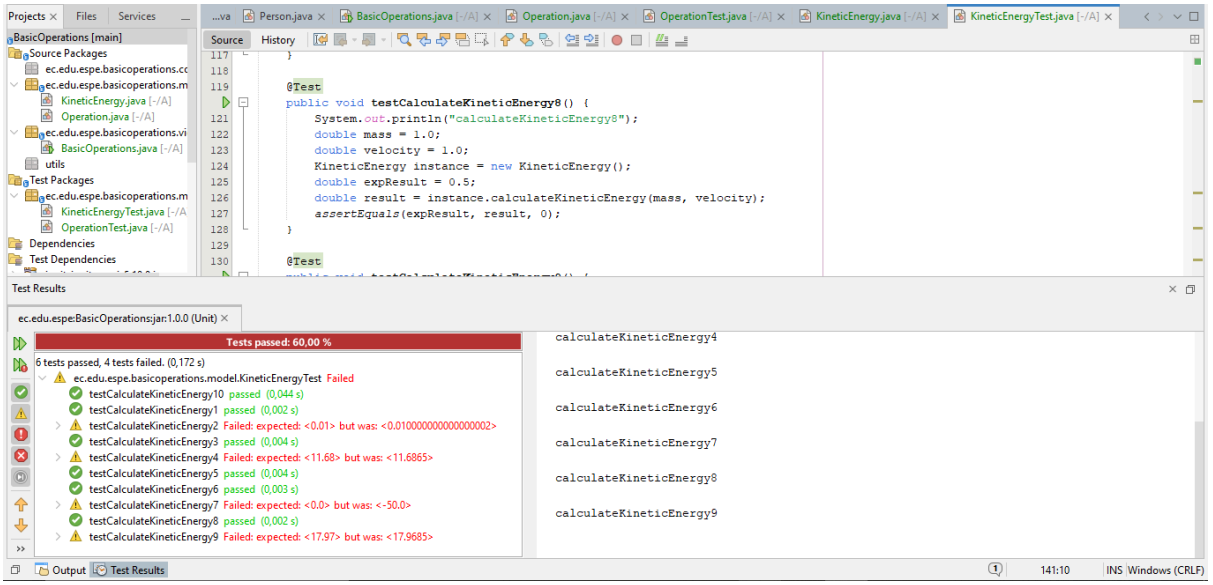
/**
 *
 * @author Jesus Garcia, @ESPE
 */
public class KineticEnergy {

    public double calculateKineticEnergy(double mass, double velocity) {

        return 0.5 * mass * (velocity * velocity);
    }

}
```

Unit Tests



```java
    }

    @Test
    public void testCalculateKineticEnergy8() {
        System.out.println("calculateKineticEnergy8");
        double mass = 1.0;
        double velocity = 1.0;
        KineticEnergy instance = new KineticEnergy();
        double expResult = 0.5;
        double result = instance.calculateKineticEnergy(mass, velocity);
        assertEquals(expResult, result, 0);
    }

    @Test
```

Test Results

ec.edu.espe:BasicOperations:jar:1.0.0 (Unit)

Tests passed: 60,00 %

6 tests passed, 4 tests failed. (0,172 s)

- ec.edu.espe.basicoperations.model.KineticEnergyTest Failed
  - testCalculateKineticEnergy10 passed (0,044 s)
  - testCalculateKineticEnergy1 passed (0,002 s)
  - testCalculateKineticEnergy2 Failed: expected: <0.01> but was: <0.010000000000000002>
  - testCalculateKineticEnergy3 passed (0,004 s)
  - testCalculateKineticEnergy4 Failed: expected: <11.68> but was: <11.6865>
  - testCalculateKineticEnergy5 passed (0,004 s)
  - testCalculateKineticEnergy6 passed (0,003 s)
  - testCalculateKineticEnergy7 Failed: expected: <0.0> but was: <-50.0>
  - testCalculateKineticEnergy8 passed (0,002 s)
  - testCalculateKineticEnergy9 Failed: expected: <17.97> but was: <17.9685>

```
calculateKineticEnergy4

calculateKineticEnergy5

calculateKineticEnergy6

calculateKineticEnergy7

calculateKineticEnergy8

calculateKineticEnergy9
```